

Lògica en la Informàtica

Deducció en Lògica de Primer Ordre (LPO)

José Miguel Rivero

Dept. Ciències de la Computació
Universitat Politècnica de Catalunya (UPC)

Tardor 2025



El material utilitzat en aquesta presentació ha estat extret del elaborat pel professor Robert Nieuwenhuis (Dept. CS, UPC) per l'assignatura *Lògica en la Informàtica* de la FIB.

En particular, del llibre *Lógica para informáticos* - Farré, R. [et al.], Marcombo, 2011. ISBN: 9788426716941.



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

A: All people that have electric cars are ecologists.

B: If someone has a grandmother, then that someone has a mother whose mother is that grandmother.

C: A person is an ecologist if his/her mother is an ecologist.

D: Mary is John's grandmother.

E: Mary has an electric car.

F: John is an ecologist.



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

A: All people that have electric cars are ecologists.

B: If someone has a grandmother, then that someone has a mother whose mother is that grandmother.

C: A person is an ecologist if his/her mother is an ecologist.

D: Mary is John's grandmother.

E: Mary has an electric car.

F: John is an ecologist.

Resposta (predicats):

$hasEcar(x)$ $\equiv$ "x has an electric car"

$isEcologist(x)$ $\equiv$ "x is an ecologist"

$mother(x, y)$ $\equiv$ "y is the mother of x"

$grandma(x, y)$ $\equiv$ "y is the grandmother of x"



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

A: All people that have electric cars are ecologists.

$hasEcar(x) \equiv$ "x has an electric car"

$isEcologist(x) \equiv$ "x is an ecologist"

$mother(x, y) \equiv$ "y is the mother of x"

$grandma(x, y) \equiv$ "y is the grandmother of x"

A: $\forall x (hasEcar(x) \rightarrow isEcologist(x))$

A $\neg hasEcar(x) \vee isEcologist(x)$



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

B: If someone has a grandmother, then that someone has a mother whose mother is that grandmother.

$hasEcar(x) \equiv$ "x has an electric car"

$isEcologist(x) \equiv$ "x is an ecologist"

$mother(x, y) \equiv$ "y is the mother of x"

$grandma(x, y) \equiv$ "y is the grandmother of x"

NO és correcte:

$\forall x (\exists y (grandma(x, y) \rightarrow \exists z (mother(x, z) \wedge mother(z, y))))$

$\forall x (\exists y (\neg grandma(x, y) \vee \exists z (mother(x, z) \wedge mother(z, y))))$



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

B: If someone has a grandmother, then that someone has a mother whose mother is that grandmother.

NO és correcte:

$\forall x (\exists y (\neg grandma(x, y) \vee \exists z (mother(x, z) \wedge mother(z, y))))$

Aquesta formalització del llenguatge natural no és adequada:

si tenim una situació I amb persones $D_I = \{p1, p2, avia\}$ i on

l'àvia de $p1$ es $avia$: $grandma_I(p1, avia) = 1$

i on tota la resta és fals (ningú és mare de ningú, etc.) llavors

I satisfà la fórmula, perquè $\forall x \exists y \neg grandma(x, y)$. De fet, amb aquesta formalització no és possible obtenir la clàusula buida.



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

B: If someone has a grandmother, then that someone has a mother whose mother is that grandmother.

El que sí és correcte és:

$$\forall x \forall y (\text{grandma}(x, y) \rightarrow \exists z (\text{mother}(x, z) \wedge \text{mother}(z, y)))$$

► Elim. $\rightarrow$

$$\forall x \forall y (\neg \text{grandma}(x, y) \vee \exists z (\text{mother}(x, z) \wedge \text{mother}(z, y)))$$

► Skolem:

$$\forall x \forall y (\neg \text{grandma}(x, y) \vee (\text{mother}(x, f_z(x, y)) \wedge \text{mother}(f_z(x, y), y)))$$

► Distrib:

$$B1 \quad \neg \text{grandma}(x, y) \vee \text{mother}(x, f_z(x, y))$$

$$B2 \quad \neg \text{grandma}(x, y) \vee \text{mother}(f_z(x, y), y)$$



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

C: A person is an ecologist if his/her mother is an ecologist.

$hasEcar(x) \equiv$ "x has an electric car"

$isEcologist(x) \equiv$ "x is an ecologist"

$mother(x, y) \equiv$ "y is the mother of x"

$grandma(x, y) \equiv$ "y is the grandmother of x"

C: $\forall x (\exists y (mother(x, y) \wedge isEcologist(y)) \rightarrow isEcologist(x))$

$\forall x (\neg \exists y (mother(x, y) \wedge isEcologist(y)) \vee isEcologist(x))$

$\forall x (\forall y \neg (mother(x, y) \wedge isEcologist(y)) \vee isEcologist(x))$

$\forall x (\forall y (\neg mother(x, y) \vee \neg isEcologist(y)) \vee isEcologist(x))$

C $\neg mother(x, y) \vee \neg isEcologist(y) \vee isEcologist(x)$



6) (4 points) Formalize and prove by resolution that sentence F is a logical consequence of the first five:

$hasEcar(x) \equiv$ "x has an electric car"

$isEcologist(x) \equiv$ "x is an ecologist"

$mother(x, y) \equiv$ "y is the mother of x"

$grandma(x, y) \equiv$ "y is the grandmother of x"

D: Mary is John's grandmother.

D $grandma(john, mary)$

E: Mary has an electric car.

E $hasEcar(mary)$

F: John is an ecologist.

$\neg$ F: John is not an ecologist.

$\neg$ F $\neg isEcologist(john)$



Examen final de 2020 tardor. Exercici 6

Volem demostrar que $A \wedge B \wedge C \wedge D \wedge E \models F$.

I això passa ssi $A \wedge B \wedge C \wedge D \wedge E \wedge \neg F$ és insatisfactible.

A $\neg hasEcar(x) \vee isEcologist(x)$

B1 $\neg grandma(x, y) \vee mother(x, f_z(x, y))$

B2 $\neg grandma(x, y) \vee mother(f_z(x, y), y)$

C $\neg mother(x, y) \vee \neg isEcologist(y) \vee isEcologist(x)$

D $grandma(john, mary)$

E $hasEcar(mary)$

$\neg F$ $\neg isEcologist(john)$

He d'obtenir la $\square$ mitjançant resolució a partir d'aquestes 7 clàusules.



Examen final de 2020 tardor. Exercici 6

- A $\neg \text{hasEcar}(x) \vee \text{isEcologist}(x)$
- B1 $\neg \text{grandma}(x, y) \vee \text{mother}(x, f_z(x, y))$
- B2 $\neg \text{grandma}(x, y) \vee \text{mother}(f_z(x, y), y)$
- C $\neg \text{mother}(x, y) \vee \neg \text{isEcologist}(y) \vee \text{isEcologist}(x)$
- D $\text{grandma}(\text{john}, \text{mary})$
- E $\text{hasEcar}(\text{mary})$
- $\neg F$ $\neg \text{isEcologist}(\text{john})$

res entre mgu

- E+A $\{x = \text{mary}\}$ obtenim:
 - 1. $\text{isEcologist}(\text{mary})$
- D+B1 $\{x = \text{john}, y = \text{mary}\}$ obtenim:
 - 2. $\text{mother}(\text{john}, f_z(\text{john}, \text{mary}))$
- D+B2 $\{x = \text{john}, y = \text{mary}\}$ obtenim:
 - 3. $\text{mother}(f_z(\text{john}, \text{mary}), \text{mary})$
- 2+C $\{x = \text{john}, y = f_z(\text{john}, \text{mary})\}$ obtenim:
 - 4. $\neg \text{isEcologist}(f_z(\text{john}, \text{mary})) \vee \text{isEcologist}(\text{john})$
- 4+ $\neg F$ $\{ \}$ obtenim:
 - 5. $\neg \text{isEcologist}(f_z(\text{john}, \text{mary}))$



Examen final de 2020 tardor. Exercici 6

- A $\neg \text{hasEcar}(x) \vee \text{isEcologist}(x)$
- B1 $\neg \text{grandma}(x, y) \vee \text{mother}(x, f_z(x, y))$
- B2 $\neg \text{grandma}(x, y) \vee \text{mother}(f_z(x, y), y)$
- C $\neg \text{mother}(x, y) \vee \neg \text{isEcologist}(y) \vee \text{isEcologist}(x)$
- D $\text{grandma}(\text{john}, \text{mary})$
- E $\text{hasEcar}(\text{mary})$
- $\neg F$ $\neg \text{isEcologist}(\text{john})$

res entre mgu

- 1. $\text{isEcologist}(\text{mary})$
- 2. $\text{mother}(\text{john}, f_z(\text{john}, \text{mary}))$
- 3. $\text{mother}(f_z(\text{john}, \text{mary}), \text{mary})$
- 4. $\neg \text{isEcologist}(f_z(\text{john}, \text{mary})) \vee \text{isEcologist}(\text{john})$
- 5. $\neg \text{isEcologist}(f_z(\text{john}, \text{mary}))$
3+C $\{x = f_z(\text{john}, \text{mary}), y = \text{mary}\}$ obtenim:
- 6. $\neg \text{isEcologist}(\text{mary}) \vee \text{isEcologist}(f_z(\text{john}, \text{mary}))$
6+5 $\{ \}$ obtenim:
- 7. $\neg \text{isEcologist}(\text{mary})$
1+7 $\{ \}$ obtenim:
- 8. $\square$



Per a estudiar teoria de LI:

- repassa els materials que hem estudiat.
- FÉS ELS EXÀMENS PENJATS, començant pels últims, cap als anteriors, treballant sempre primer l'enunciat SENSE resoldre, i després l'examen resolt.
- continua fent els exercicis del tema 5.

